FPGA Experiment Report

Experiment 6 — VGA Protocol (Screen Output)

Task: FPGA 6 — VGA interface for screen output

Student 1: Mahmood Stitia **ID:** 214032682

Student 2: Saleh Khalil **ID:** 213310485

Part 1 — Status update

The table below lists the submitted files and their status. The status is one of *Done* (module and simulation written and passing), *Partially done* (started but not fully working — explained in the notes), or *Not started*.

`File / Step`	Status	Notes / comments
`VGA_Interface.v`		
`VGA_Interface_tb.v`		
`Drawer.v`		
`Debouncer.v`		
`VGA_Top.v`		
`task2.xdc`		

Issues

Any errors present at the time of submission are described here (per module / simulation, with what the error means).

- `VGA_Interface.v`:
- `VGA_Interface_tb.v`:
- `Drawer.v`:
- `Debouncer.v`:
- `VGA_Top.v`:
- `task2.xdc`:

Part 2 — Simulations and answers

Simulation — VGA_Interface

The design targets the **800**$\times$**600** @ $72\,\mathrm{Hz}$ VGA mode driven by a $50\,\mathrm{MHz}$ pixel clock, generated by dividing the board's $100\,\mathrm{MHz}$ system clock by two. Two counters track the beam position: `h_count` runs 0 to 1039 (1040 pixel-clocks per line) and `v_count` runs 0 to 665 (666 lines per frame). Only the 800×600 window is visible; outside it the outputs are blanked to black.

Simulated Scenario

1. **Reset and counter roll-over.**
2. **Sync-pulse timing.**
3. **Visible vs. blanking colour gating.**

Pixel-clock and Blanking Considerations

[paste screenshot here]

*Full `VGA_Interface_tb` simulation with the TCL Console message **[test result here]**.*

Signals (top to bottom in the waveform):

- `clk` - $100\,\mathrm{MHz}$ system clock (test-bench driven).
- `rstn` - active-low reset.
- `pclk` - internal $50\,\mathrm{MHz}$ pixel clock (clk divided by two).
- `pixel_color[11:0]` - input colour word {R,G,B} from the Drawer.
- `XCoord[10:0]` / `YCoord[10:0]` - current pixel position.
- `Hsync` / `Vsync` - sync outputs (positive polarity).
- `vgaRed/Green/Blue[3:0]` - colour outputs (forced to 0 in blanking).

[paste screenshot here]

Annotated close-up of the same simulation. the sync edges and the visible-to-blanking transition.

Elaborated Design

Top module

[paste screenshot here]

RTL-elaborated schematic of `VGA_Top`: `vga_if` (`VGA_Interface`) and `drawer` (`Drawer`), with the `reset` push-button inverted into `rstn`.

VGA_Interface

[paste screenshot here]

RTL-elaborated schematic of the VGA_Interface.

Drawer

[paste screenshot here]

RTL-elaborated schematic of the Drawer.

Implementation

Synthesis schematic

[paste screenshot here]

Post-synthesis schematic.

Project summary

[paste screenshot here]

Post-implementation project summary.

Timing summary

[paste screenshot here]

Design Timing Summary after implementation.

Critical Path

[paste screenshot here]

Intra-Clock Paths report (worst path, slack, logic levels and delay).

Device

[paste screenshot here]

Device (floorplan) view of the placed-and-routed design on the Artix-7 (`XC7A35T`).

Problems we faced

[paste screenshot here]

The full system on hardware: the Basys-3 board driving the monitor over VGA, showing the movable rectangle (size/colour controlled by the switches and buttons).

Appendix — Verilog Source Code

The full source of the submitted files is listed below. Paste the code of each file inside its block.

VGA_Interface.v

```
// paste VGA_Interface.v here
```

VGA_Interface_tb.v

```
// paste VGA_Interface_tb.v here
```

Drawer.v

```
// paste Drawer.v here
```

Debouncer.v

```
// paste Debouncer.v here
```

VGA_Top.v

```
// paste VGA_Top.v here
```

task2.xdc

```
# paste task2.xdc here
```